CE 310

The Normal Distribution

Week 5 · Session 1 · The Bell Curve in Engineering

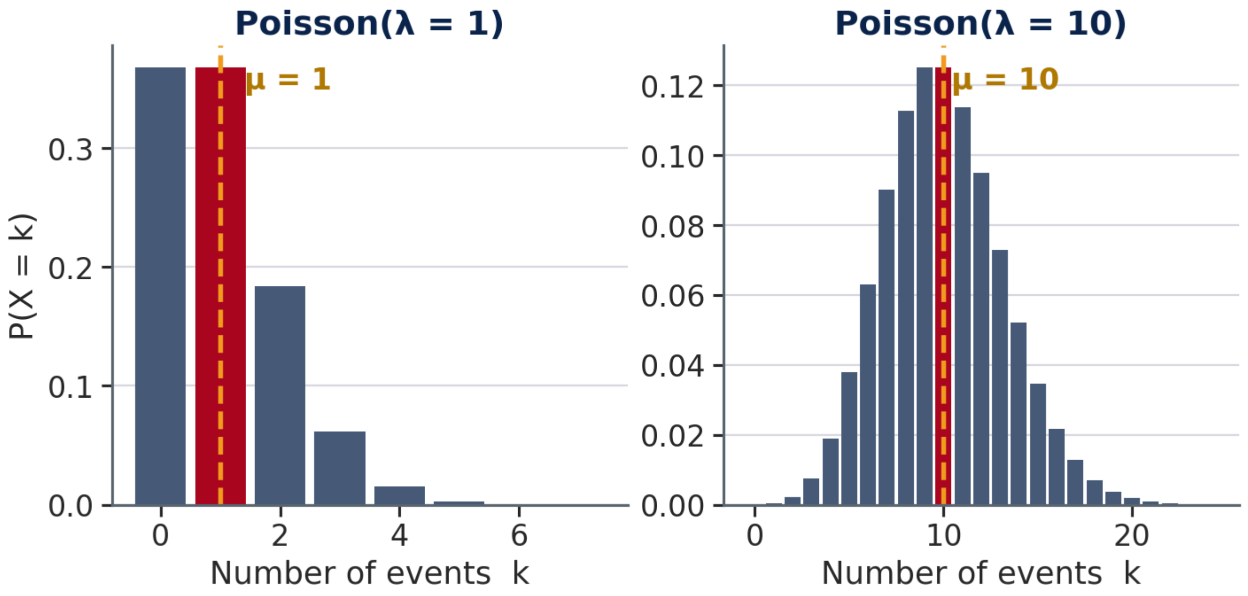
University of Arizona · Dr. Wooyoung Jung · Fall 2026

Part 1

Where the Normal Distribution Comes From

From Week 4 — The Question We Left Open

In Week 4 we built the Poisson model for discrete event counts. At large λ, something interesting happens to its shape:



As λ grows, the Poisson PMF:

- Shifts right (mean = λ)
- **Becomes increasingly bell-shaped**

At λ = 1 events are rare and the shape skews sharply right. At λ = 10 it already resembles a bell.

> 💡 At large λ, Poisson approaches a famous continuous distribution. Today we name it and build the tools to use it.

Try It — Sum of Dice

3 minutes · no computers · just arithmetic

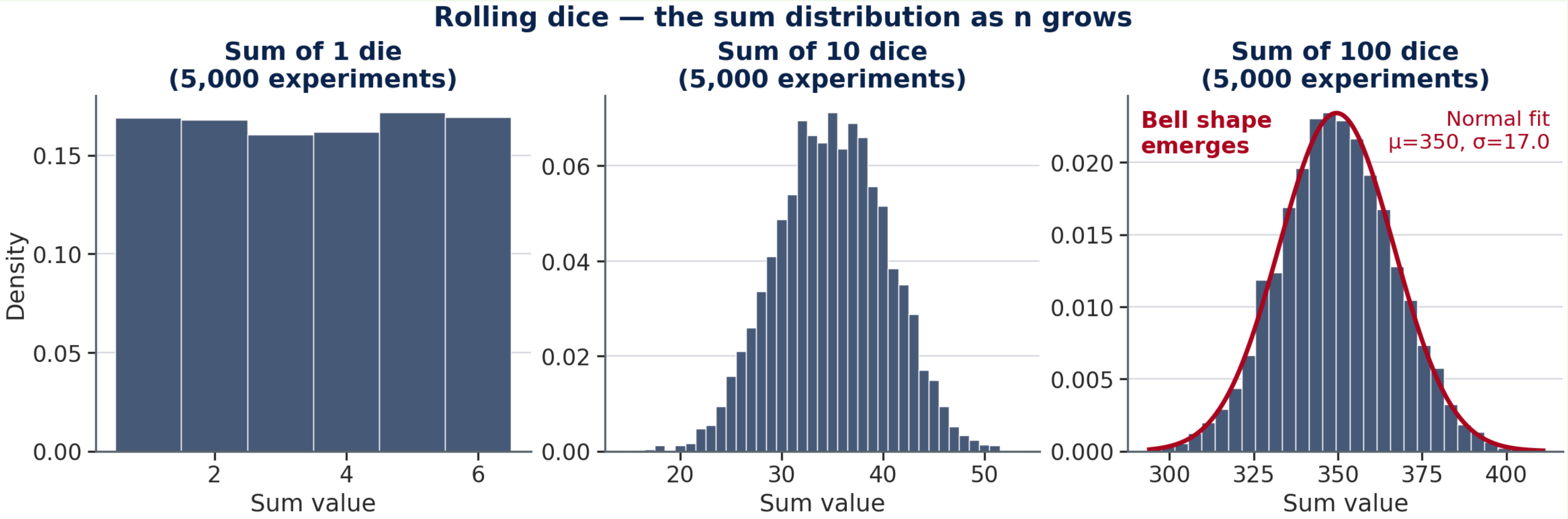
You need: a die (physical or mental — pick a random number 1 through 6 each roll).

Round 1: Roll once. Write down your number.

Round 2: Roll **10 times**. Add all 10 numbers together. Write down the sum.

> When everyone has a sum: call out your Round 2 answer. What range do you expect? Where do most answers cluster?

What Happens When You Add Dice



Each panel shows 5,000 simulated experiments. No formula was imposed — the bell shape **emerged** from repeated addition. The red curve on the right panel is a Normal distribution fitted to the data.

The Central Limit Theorem

Theorem: If $X_1, X_2, \ldots, X_n$ are independent random variables with the same distribution (any shape), mean μ, and standard deviation σ, then as n grows large:

$$\frac{X_1 + X_2 + \cdots + X_n}{n} \longrightarrow N\left(\mu, \ \frac{\sigma^2}{n}\right)$$

What this says: No matter what distribution each individual piece follows — uniform dice, exponential service times, Poisson counts — their *average* (or sum) approaches a Normal distribution as more pieces are added.

💡 Three things carry through to the bell: the mean μ, the spread σ, and the number of pieces n. The shape of each individual piece does not matter.

Why Normal Appears in Engineering

When many independent quantities are added, their sum tends toward a bell shape — regardless of the shape of each individual piece.

Quantity	What it sums
Concrete f'c	Water ratio, aggregate, curing temp, placement
Traffic speed	Reaction time, speed selection, vehicle response, road geometry
Structural dead load	Self-weight of beams, slab, finishes, MEP
Annual rainfall	Many independent storm events

💡 Whenever an engineering quantity is the result of many small, independent influences — manufacturing variation, human behavior, environmental fluctuation — the Normal distribution is the natural starting model.

Today's Dataset — Four Normal Variables

Four engineering quantities, all well-modeled by $X \sim N(\mu,\, \sigma^2)$:

Variable	μ	σ	Design application
Concrete f'c (psi)	4,632	344	ACI 318: 10th percentile ≥ specified f'c
Traffic speed (mph)	54.4	7.8	85th percentile → posted speed limit check
Outdoor temp (°F)	68.7	16.9	ASHRAE 90.1: 99th percentile → equipment sizing
Steel yield (ksi)	50.3	3.2	LRFD: reliability-based load and resistance factors

> 💡 This week's in-class exercise uses the first two rows — concrete QC and traffic speed. You will compute μ and σ from data, then run the curve in both directions to answer the design questions in the last column.

Today's Tools — Five Concepts

What we need	Tool
Characterize a distribution	μ, σ, Normal model
$P(X \leq \text{threshold})$	Cumulative distribution (CDF)
$P(X > \text{threshold})$	Exceedance = 1 – CDF
$P(a < X < b)$	Range probability
Design-condition value	Inverse CDF

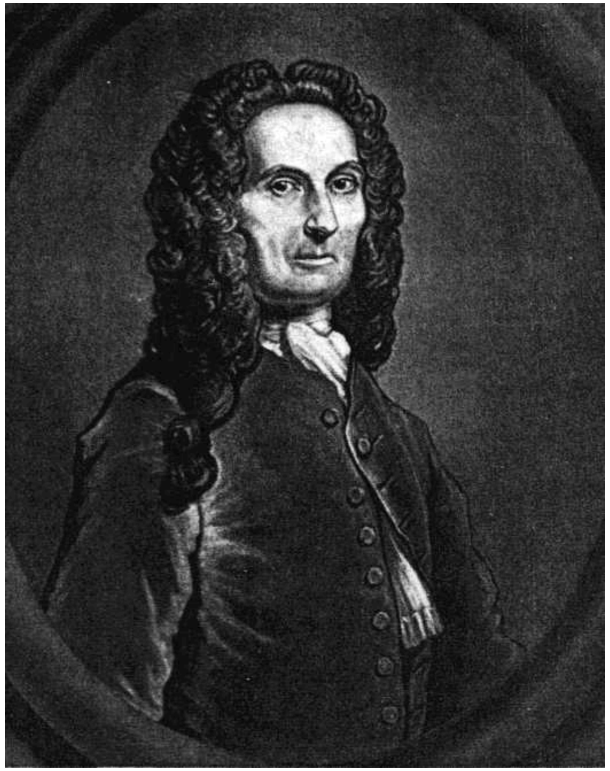
Dataset: `Week5_CE_Measurements.csv` — 500 records · concrete QC specimens + traffic speed observations

Columns: Supplier (col B), TimeOfDay (col C), fc_psi (col D), Speed_mph (col E), CostPerCY (col F)

Part 2

The Normal Model: μ and σ

The Normal Distribution — de Moivre, 1733



Abraham de Moivre (1667–1754)

Abraham de Moivre — French Huguenot refugee, London actuary

Working on gambling and insurance problems, de Moivre approximated the binomial distribution for large n. In 1733, he found the bell curve was its limiting shape — a **mathematical approximation tool**, not a model of physical reality (*Doctrine of Chances*, 1738).

> 💡 A calculation shortcut for coin flips — not yet engineering reliability's foundation.

The Normal Distribution — Gauss, 1809

Carl Friedrich Gauss (1777–1855)

Carl Friedrich Gauss — German mathematician, astronomer

In 1809, Gauss was tracking Ceres — an asteroid observed 41 days before vanishing behind the Sun. Combining noisy telescope readings to predict its orbit, he asked *what error distribution makes the mean the best estimator?* The answer: the bell curve, still called the **Gaussian distribution**.

> 💡 Gauss was combining noisy measurements to track a moving object — the same math now describes concrete strength and load variability across CE.

The Normal Distribution — Quetelet, 1835

Adolphe Quetelet (1796–1874)

Adolphe Quetelet — Belgian statistician, astronomer

In 1835, Quetelet measured human physical characteristics — height, weight, chest circumference of Belgian soldiers — and found that the same bell curve described human variation. He introduced the concept of the **"average man"** (*l'homme moyen*).

This is when the name **Normal** emerged: the distribution of typical, natural variation. Engineers say **Gaussian**; statisticians say **Normal**. Same formula: $X \sim N(\mu, \sigma^2)$.

What the Normal Distribution Is

A random variable X follows a **Normal distribution** with mean μ and standard deviation σ. We write $X \sim N(\mu,\, \sigma^2)$. Its PDF:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Three properties that matter for engineers:

1. **Symmetric** around μ — distribution is identical on both sides of the mean
2. **Bell-shaped** — most probability near the center, tailing off toward both extremes
3. **Fully described by two parameters** — once you know μ and σ, you know everything
4. **σ is a distance you can see** — the curve bends *downward* between μ – σ and μ + σ, and *upward* outside it. Those two turning points sit exactly at μ ± σ

The Normal PDF is never negative and integrates to exactly 1 over all real values.

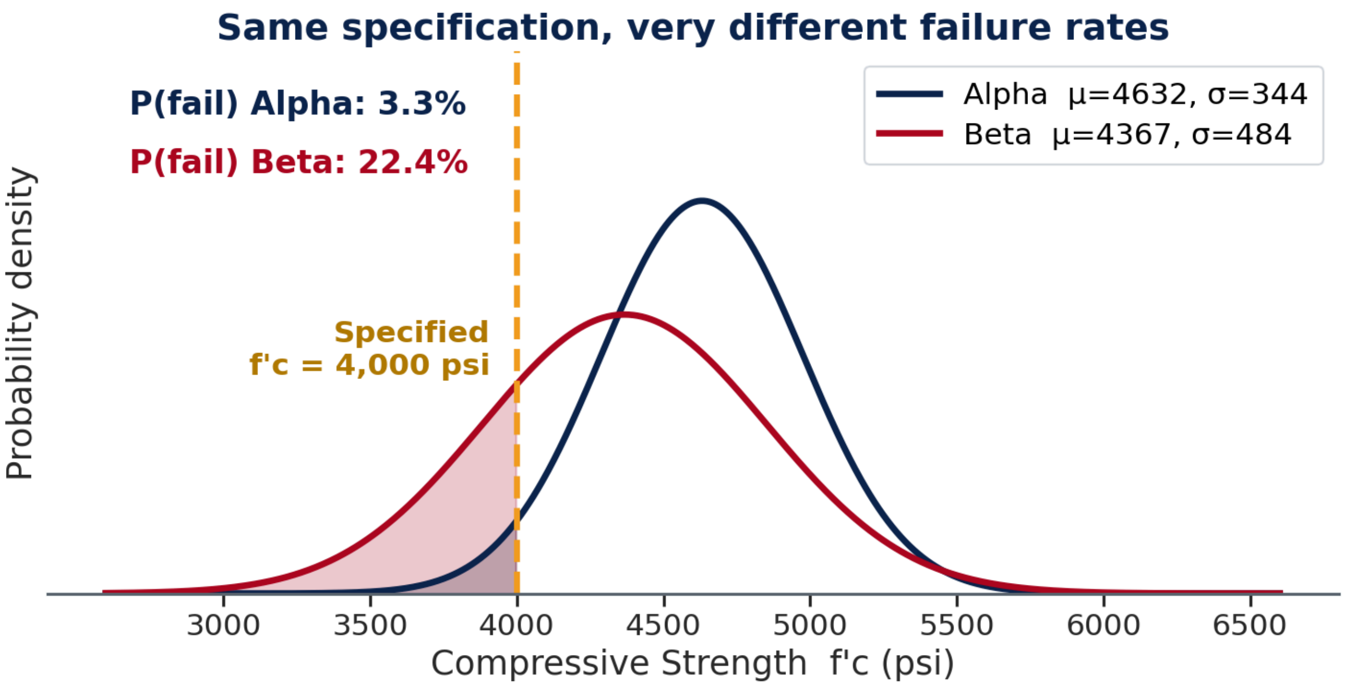
What μ and σ Control

μ shifts the distribution left or right.

- Alpha (μ = 4,632 psi) centered higher · Beta (μ = 4,367 psi) centered lower

σ controls the width — same shape, different spread.

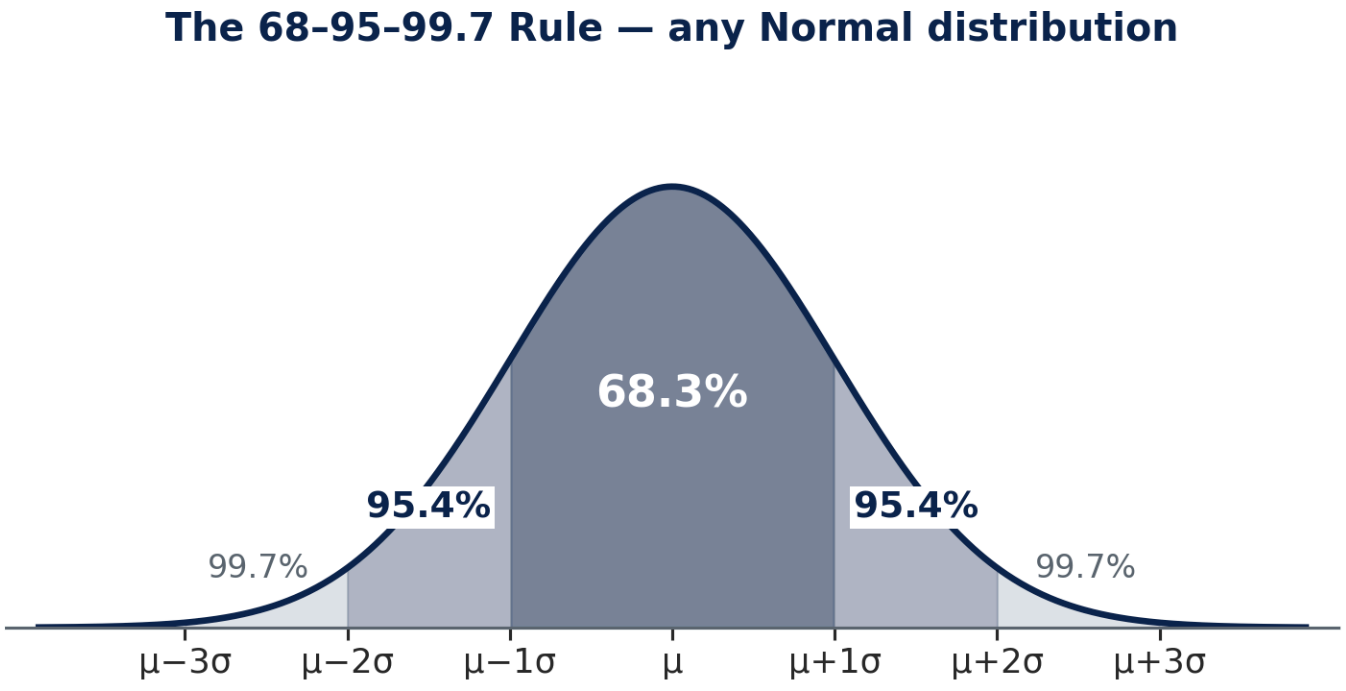
Supplier	σ (psi)	Interpretation
Alpha	**344**	Tighter QC — consistent mix
Beta	**484**	Looser QC — more variability
Ratio	**1.41×**	Beta is 41% more variable



💡 Same target f'c = 4,000 psi — failure rates differ dramatically because of σ alone.

The 68–95–99.7 Rule

For **any** Normal distribution — memorize once, apply everywhere:

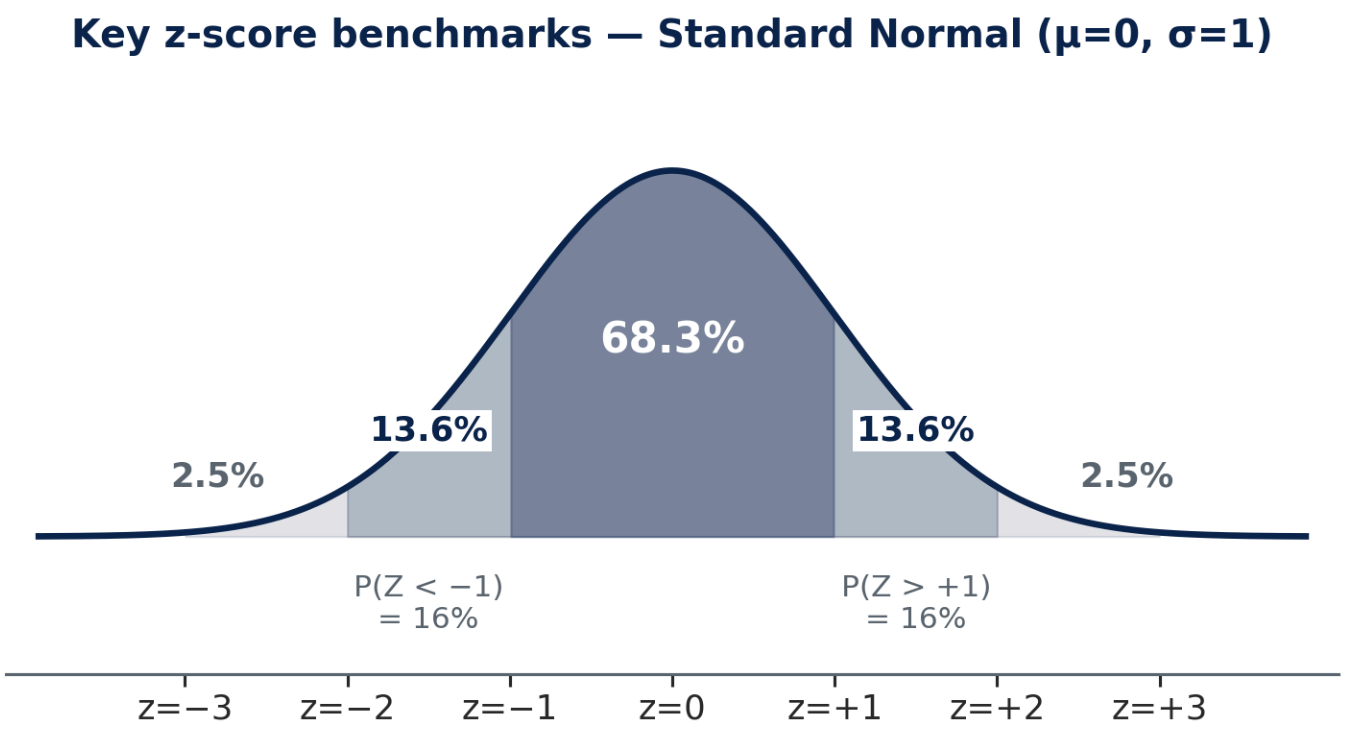


Range	Probability
$\mu \pm 1\sigma$	**68.3%**
$\mu \pm 2\sigma$	**95.4%**
$\mu \pm 3\sigma$	**99.7%**

Concrete example (μ = 4,496, σ = 442 psi):

- $\mu \pm 1\sigma \rightarrow$ 4,054 to 4,938 psi
- $\mu \pm 2\sigma \rightarrow$ 3,612 to 5,380 psi
- $\mu \pm 3\sigma \rightarrow$ 3,170 to 5,822 psi

68–95–99.7 — Reading the Boundaries



For **non-integer z**, bracket between known benchmarks:

z	P(Z < z)
–2	**2.5%**
–1	**16%**
+1	**84%**
+2	**97.5%**

Example: Alpha spec limit → z = –1.84, between –2 and –1, so 2.5% < P(f_c < 4000) < 16%.

Estimate: **~3–6%**. The CDF gives the exact value.

The z-Score — Standardizing Any Normal

The **z-score** converts any value from any Normal distribution into standard units — how many standard deviations the value sits above or below the mean:

$$z = \frac{x - \mu}{\sigma}$$

z	Meaning
z = 0	Exactly at the mean
z = +1	One σ above mean
z = −2	Two σ below mean
z = +1.036	85th percentile
z = −1.282	10th percentile

Concrete example — all specimens (μ=4,496, σ=442):

- Spec limit (4,000 psi): $z = \frac{4000 - 4496}{442} = -1.12$
- High-strength (5,500 psi): $z = \frac{5500 - 4496}{442} = +2.27$
- Traffic design speed (mph) uses the same formula — just different units

z-Score Practice — Concrete QC

3 minutes · think before you compute

Context: Supplier Alpha — 243 specimens, μ = 4,632 psi, σ = 344 psi. Specified f'c = 4,000 psi.

Question A: Compute the z-score for the specification limit (x = 4,000 psi).

Question B: Using the 68–95–99.7 rule, estimate P(fc < 4,000 psi). Does Alpha meet ACI's 10% limit?

Question C: A specimen tests at 5,200 psi. What is its z-score? Is this unusual?

> Compare with your neighbor.

✅ z-Score Practice — Answers

Supplier Alpha: $\mu = 4{,}632$, $\sigma = 344$ psi

Question A — z-score for spec limit (4,000 psi):

$$z = \frac{4000 - 4632}{344} = \frac{-632}{344} = -1.84$$

Question B — Estimate P(fc < 4,000 psi):

$z = -1.84$ sits between –2 ($P \approx 2.5\%$) and –1 ($P \approx 16\%$). Rough estimate: **3–6%**. This is below ACI's 10% limit → Alpha appears compliant. Exact answer (Part 3): **3.3%**.

Question C — z-score for 5,200 psi specimen:

$$z = \frac{5200 - 4632}{344} = +1.65$$

top ~5% of specimens — strong result, not extreme

✅ z-Score Practice — Key Takeaway

> 💡 The 68–95–99.7 rule gives a fast estimate at integer z-values. For non-integer z (like –1.84), it brackets the answer. The CDF gives the precise value — which is what the in-class exercise requires.

- **Bracket:** $z = -1.84$ → between –2 (2.5%) and –1 (16%) → estimate **3–6%**
- **Exact:** the Normal CDF at 4,000 psi ($\mu = 4{,}632$, $\sigma = 344$) = **3.3%** — inside the bracket ✓
- **What z = –1.282 means:** $\mu - 1.282\sigma$ is the 10th percentile — the ACI 318 compliance threshold (Part 4)

Part 3

From σ to Probability

What You Are Assuming When You Fit a Curve

Everything from here treats the measurements **as if** they were Normal — as if two numbers, μ and σ, described them completely.

What the assumption buys you

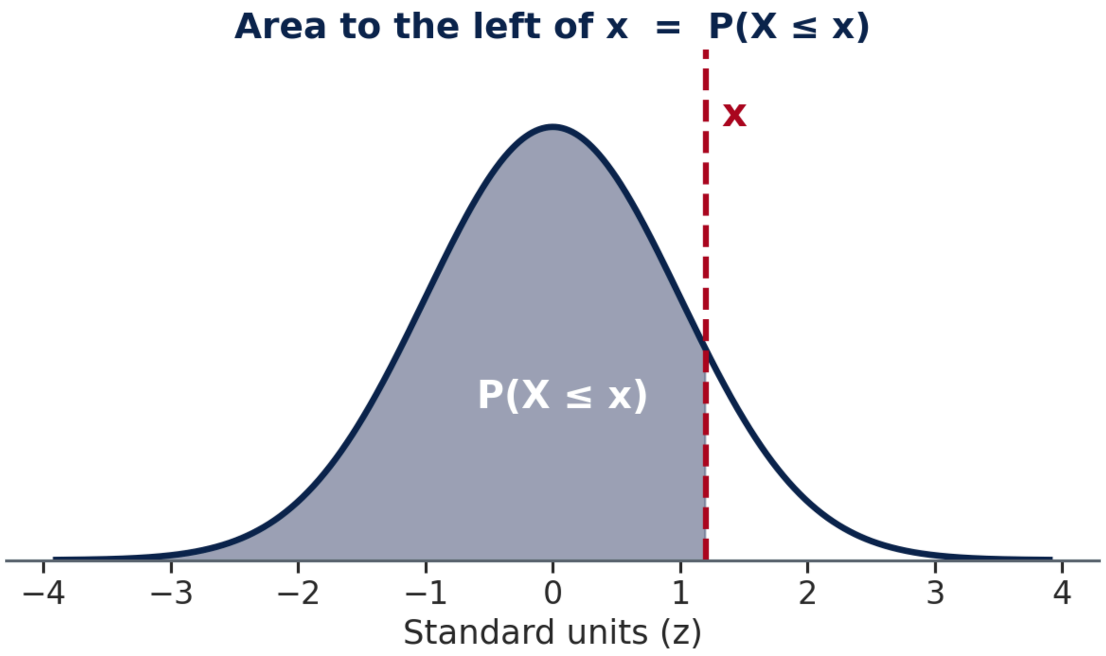
- Any probability, at any threshold, from just μ and σ
- Answers *beyond* your data — a 1-in-50-year load from 30 years of readings
- One method that transfers across every domain

What it costs

- The answer is only as good as the fit
- A skewed variable forced into a symmetric curve goes wrong **where the curve is thinnest — the tails**
- And design thresholds live in the tails

> 💡 Every number you compute this week inherits this assumption. **Week 6 is where we test it** — by counting the real records and seeing how far the curve was off.

The Bell Curve — Visualizing $P(X \le x)$



Total area = 1. The CDF returns the shaded **fraction** to the left of x.

Region	What it means
Left of x	$P(X \le x)$
Right of x	$P(X > x) = 1 - \text{left}$
Between a and b	$P(a < X < b)$

Moving x right: shaded area increases toward 1.
Moving x left: shaded area decreases toward 0.
At $x = \mu$: exactly **50%** shaded (symmetric distribution).

There is no formula for that area — the bell curve has no elementary antiderivative. Engineers read printed tables for two centuries; every tool now ships a function for it.

From a Value to a Probability

Give the curve a value, and it returns the fraction of the population below it. That is the cumulative distribution function — the CDF.

Two different things you can read off a bell curve, and only one is a probability:

What you read	For concrete at 4,000 psi	Is it a probability?
The **area** to the left of x	**0.131** — 13.1% of specimens fall below	✓ a fraction between 0 and 1
The **height** of the curve at x	0.000481 **per psi**	✗ a density, with units; it can exceed 1

Concrete, all specimens: μ = 4,496 psi, σ = 442 psi.

The area to the left of the 4,000 psi specification is **13.1%** — about one cylinder in eight.

CDF definition: $F(x) = P(X \le x)$ — the area under the bell curve from $-\infty$ to x.

The area is an integral, and the spreadsheet evaluates it for you. You never integrate by hand in this course.

Three Probability Patterns — Learn These Three

Every Normal probability question reduces to one of three patterns:

Question	Pattern
$P(X \leq x)$ — left tail	the CDF, read directly
$P(X > x)$ — right tail	1 – CDF
$P(a < X < b)$ — between	CDF(b) – CDF(a)

Every one of them is *an area under the curve*, or one area subtracted from another. There is no fourth case.

Three Patterns — Vocabulary Map

Map the question's wording to the pattern before computing:

If the question says…	Use this pattern
"less than," "below," "under," "at most"	Left tail — CDF directly
"more than," "above," "exceeds," "at least"	Right tail — 1 – CDF
"between," "within," "from X to Y"	Middle — CDF(b) – CDF(a)

> ⚠️ Read the wording before you reach for a tool. Almost every wrong answer on this material comes from picking the wrong pattern, not from computing the area badly.

Quick Check — Three Probability Patterns

3 minutes · patterns first, numbers second

Question A: Concrete specimens, $\mu = 4{,}496$, $\sigma = 442$ psi. An inspector asks: "What fraction of specimens fall between 4,200 and 5,000 psi?" Name the pattern that applies and write the general expression before touching Excel.

Question B: Traffic speeds, $\mu = 54.4$ mph, $\sigma = 7.8$ mph. Using only the 68–95–99.7 rule and z-scores — no Excel — estimate $P(\text{Speed} > 70 \text{ mph})$. Is 70 mph unusual on this road?

Question C: For any Normal distribution, what fraction of the population falls below the mean, and why? And what is the *height* of the curve at the mean — is that a probability?

> Compare answers with your neighbor before we confirm.

✅ Quick Check — Answers

Question A: Pattern = **CDF(b) – CDF(a)** — the "between" pattern. Take the area below 5,000 and subtract the area below 4,200. Result ≈ **56.7%**.

Question B: $z = (70 - 54.4)/7.8 =$ **+2.0** → right tail beyond $+2\sigma$ → **≈ 2.5%**. Yes, 70 mph is unusually fast — only 1 in 40 vehicles reach that speed.

Question C: 0.50 — exactly half the distribution lies below the mean, because the Normal is symmetric. The height at the mean is the curve's peak, $\approx 0.399/\sigma$ — **not a probability**; it carries units of per-psi (or per-mph) and can exceed 1 for a narrow distribution.

✅ Quick Check — Important Reminder

⚠️ Sanity-check every answer against the picture. A probability is an *area*, so it lives between 0 and 1 and has no units. If a result comes back as 0.000481 per psi, you have read the curve's height instead of its area.

- **Sanity check:** the fraction below μ is **0.50** exactly — the Normal is symmetric, so mean = median
- **Sanity check:** the fraction below $\mu + 3\sigma$ is $\approx$ **0.9987** — the 99.7% rule, seen from the left
- If a result fails either check, you have read the wrong quantity off the curve

Part 4

From Probability to Design Value

From a Probability to a Value

The design question engineers actually ask: "What strength value keeps failure below 10%?" — not "what fraction fails at a given strength?"

Geometric idea: the CDF maps x → area (probability). The inverse CDF runs it backward: given a target probability p, find the x-value on the bell curve that has exactly that area to its left.

$$\text{If } P(X \le x) = p, \quad \text{then we want the } x \text{ that makes it so.}$$

Lower-tail limit — exclude the weakest fraction:

- Keep only **10%** below the spec → the 10th percentile of the fitted curve
- ACI 318 requires this value ≥ f'c

From a Probability to a Value (continued)

Upper-tail limit — include nearly all values:

- **85%** of drivers at or below design speed → the 85th percentile of the fitted curve
- MUTCD uses this to set posted speed limits

> 💡 These are one question asked in two directions. **Value** → **fraction** checks a design you already have; **fraction** → **value** produces the design in the first place. Real problems use both: set the threshold, then check the failure rate it implies.

Design Conditions Across CE — Same Tool, Different Percentiles

Every one of these standards names a fraction and asks for the value that sits at it:

Domain	Standard	Design condition	What that tolerates
Structural (concrete)	ACI 318-19	10th percentile f'c ≥ specified	1 test cylinder in 10 below spec
Transportation	MUTCD / ITE	85th percentile speed	15% of drivers above design speed
Geotechnical	Eurocode 7	5th percentile bearing capacity	1 sample in 20 weaker than assumed
MEP / Arch	ASHRAE 90.1	99th percentile outdoor temperature	≈ 88 hours a year the equipment can't hold

Why These Percentiles?

Four standards, four different percentiles. None of them is a statistical convention — each was chosen deliberately.

Standard	Percentile
Concrete compressive strength (ACI 318)	**10th**
Highway design speed	**85th**
Geotechnical bearing capacity	**5th**
HVAC equipment sizing	**99th**

Two questions:

1. Why is concrete a *lower* tail and design speed an *upper* one?

2. What makes geotechnical more conservative than concrete?

✅ Design Conditions — Consequence, Not Statistics

The percentile in each standard reflects **consequence of failure**, not statistics. The statistics implement the decision:

- **Structural failure (10th percentile):** lower tail → keeps weak specimens out → conservative against failure
- **Design speed (85th percentile):** upper tail → captures nearly all drivers → limit matches natural traffic behavior
- **Geotechnical (5th percentile):** very conservative lower tail → soil failure is often catastrophic and irreversible
- **Equipment sizing (99th percentile):** extreme conditions are rare but costly to miss → very high upper tail

ACI 318 — The 10th Percentile Rule

The engineering requirement: When concrete leaves the plant, a fraction of test cylinders will test below the specified strength f'c. ACI 318 — the primary US concrete design code (§26.12: evaluation and acceptance of hardened concrete) — sets the tolerance:

> **No more than 10% of test cylinders may fall below the specified f'c.**

The statistical translation: this means the 10th percentile of the strength distribution must be at or above f'c.

The check: take the 10th percentile of the fitted strength curve and compare it to the specified strength.

$$\text{10th percentile of } N(\bar{f}_c,\, \sigma) \;\geq\; f'_c \quad \text{(specified strength)}$$

ACI 318 — Two Suppliers, One Specification

Specified strength f'c = 4,000 psi. Same code, same test — the difference is entirely in μ and σ.

Supplier Alpha (μ=4,632, σ=344):

10th percentile of Alpha's model (μ = 4,632, σ = 344) → **4,191 psi**

4,191 psi ≥ 4,000 psi ✓ **Compliant**

P(fail) = CDF at 4,000 psi → **3.3%** < 10%

Supplier Beta (μ=4,367, σ=484):

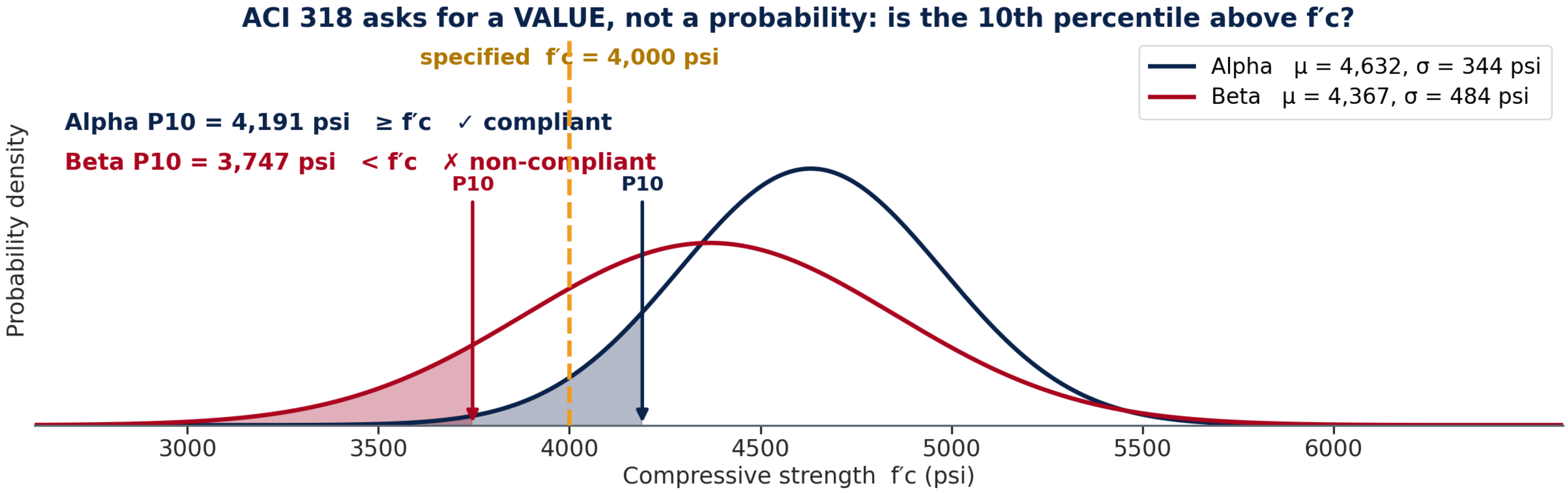
10th percentile of Beta's model (μ = 4,367, σ = 484) → **3,747 psi**

3,747 psi < 4,000 psi ✗ **Non-compliant**

P(fail) = CDF at 4,000 psi → **22.4%** >> 10%

The 10th-Percentile Rule, Drawn

Both suppliers make concrete that *averages* well above 4,000 psi. Only one of them passes.



ACI 318 — Why Beta Fails and How to Fix It

⚠️ Beta's 10th percentile (3,747 psi) falls below the 4,000 psi specification, and its 22.4% failure rate exceeds the 10% limit.

To become compliant, Beta must reduce σ, raise μ, or both:

Option	Effect	Practical approach
Raise μ	Shifts distribution right	More cement, better aggregate
Reduce σ	Narrows distribution	Tighter batching and temperature control
Both	Most effective	Standard corrective action for marginal mixes

Traffic Design Speed — Setting the Threshold

Apply the same logic to a transportation problem

A traffic engineer observes spot speeds on an urban arterial during the AM peak.

- AM Peak speeds: **μ = 52.2 mph, σ = 8.2 mph** (278 observations)
- PM Peak speeds: **μ = 57.2 mph, σ = 6.2 mph** (222 observations)
- Posted speed limit: **45 mph**

Step 1: Compute the 85th percentile AM peak speed — the standard benchmark for judging a posted limit.

Step 2: Compute P(Speed > 45 mph) for both AM and PM peaks. What fraction of drivers exceed the posted limit?

Step 3: Should the posted speed limit be raised? What does the data suggest?

> Work through Steps 1–2, then discuss Step 3 with your neighbor.

✅ Traffic Design Speed — Answers

	AM Peak	PM Peak
85th percentile speed	**60.7 mph**	**63.6 mph**
P(Speed > 45 mph)	**≈ 81%**	**≈ 99%**

85th percentile: 60.7 mph ($\mu = 52.2$, $\sigma = 8.2$) · 63.6 mph ($\mu = 57.2$, $\sigma = 6.2$)

P(Speed > 45): ≈ 81% ($\mu = 52.2$, $\sigma = 8.2$) · ≈ 99% ($\mu = 57.2$, $\sigma = 6.2$)

Step 3 — What the data suggests:

The 85th percentile speeds (61–64 mph) are 15–19 mph above the posted limit. MUTCD guidance recommends speed limits within 5–8 mph of the 85th percentile speed to reflect natural traffic behavior. A limit of 55 mph may better fit the observed distribution.

✅ Part 4 — One Tool, Four Domains

> 💡 The same move — name a fraction, read off the value that sits at it — appears in ACI 318 for concrete, MUTCD for traffic, Eurocode 7 for soil and ASHRAE 90.1 for climate. The engineering domains differ; the statistical question is identical.

- **ACI 318:** $p = 0.10$ → is the 10th percentile ≥ f'c? → compliance check
- **MUTCD:** $p = 0.85$ → 85th percentile vs. posted limit → speed study
- One direction finds the threshold, the other checks the failure rate it implies — they always appear as a pair

Part 5

Not Everything is Bell-Shaped

Construction Cost — Right-Skewed

Concrete compressive strength is approximately Normal. Unit construction cost (CostPerCY, $/CY) is not.

CostPerCY distribution:

- Mean: **$195/CY** — pulled up by high-cost jobs
- Median: **$185/CY** — the typical job
- Skewness ≈ **+1.5** — strongly right-skewed

Diagnostic check: Mean ($195) > Median ($185) → right skew → use the lognormal.

> ⚠️ Any variable bounded at zero — costs, durations, flows — is a poor fit for Normal when skewed. Normal extends to $-\infty$ and will predict physically impossible negative values in the left tail.

The Lognormal Distribution — A Preview

If a variable Y is always positive and right-skewed, and if **ln(Y)** — the natural logarithm of Y, which Excel writes `LN(Y)` — appears bell-shaped, then Y follows a **lognormal distribution**:

$$\text{If } \ln(Y) \sim N(\mu_{ln},\, \sigma^2_{ln}) \text{ then } Y \text{ is lognormal}$$

For CostPerCY in today's dataset:

Parameter	Value	Meaning
μ_ln	**5.22**	Mean of ln(CostPerCY)
σ_ln	**0.35**	Std dev of ln(CostPerCY)
Lognormal median	$e^{5.22}$ = **$185/CY**	Typical job cost
Lognormal mean	$e^{5.22+0.35^2/2}$ = **$196/CY**	Pulled up by high-cost tail

The median ($185) is below the mean ($196) — the long right tail pulls the mean up.

Normal vs Lognormal — Decision Table

Feature	Use Normal	Use Lognormal
Values	Range from $-\infty$ to $+\infty$	Always positive (> 0)
Shape	Symmetric (skew ≈ 0)	Right-skewed (skew > 0.5)
CE examples	Concrete f'c, traffic speeds, measurement errors	Construction costs, project durations, flood peaks, material loads

Normal vs Lognormal — Diagnosis and Tools

Feature	Use Normal	Use Lognormal
Diagnostic test	Mean ≈ Median	Mean > Median; ln(data) looks bell-shaped
Which curve you integrate	the Normal, on the data itself	the Normal, on ln(data)

Week 6 will go deeper: empirical CDF and Q-Q plots — visual tools for checking whether Normal or lognormal fits the data well.

Part 6

Bridges — Poisson, Correlation, and Next Week

Bridge: When Poisson Becomes Normal

Week 4 preview question answered: As λ grows large, what does the Poisson PMF approach?

At large λ, a Poisson random variable $X \sim \mathrm{Poisson}(\lambda)$ is approximately:

$$X \approx N(\mu = \lambda,\ \sigma^2 = \lambda)$$

λ	Approximation quality	Recommendation
λ < 5	Poor — strongly right-skewed	Use Poisson
5 ≤ λ ≤ 20	Moderate — Normal gives rough estimates	Either; check skewness
λ > 20	Good — Normal gives accurate probabilities	Normal is fine

Why this happens: Poisson(λ) counts the sum of λ independent rare events. By the CLT — the same mechanism as summing dice — that sum tends toward Normal as λ grows.

Bridge to Machine Learning — Feature Transformation

Week 5 tool	ML equivalent	Connection
`=LN(...)` helper column for cost	Log transform	Standard preprocessing for right-skewed features
Lognormal vs. Normal fit	Choosing the right target scale	Fit the model where the data is symmetric
Normal fit vs. the actual data	Residual diagnostics	Models assume errors are roughly Normal
`NORM.INV` for a percentile	Prediction interval	Turning a fitted model into a range, not a point

Log-transforming a right-skewed feature is one of the most common preprocessing steps in practice — and it is the `=LN(...)` column from this week's tutorial, nothing more. Fit on the raw skewed costs and a handful of large projects dominate everything; fit on the logs and every project gets comparable influence.

Three Deliverables This Week

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 8:00 AM Tuesday, Sep 29
In-Class Exercise	Sections A and B, the charts, and the memo	Work Wed + Fri · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

The exercise spans two sessions. You start it Wednesday after the tutorial, and finish it Friday. **You are not expected to finish in class** — use the session time for the parts where having me in the room actually helps, and complete the rest on your own.

> 💡 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Tuesday, Sep 29; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Looking Ahead — Week 6: Does the Data Actually Fit?

Today we **assumed** the Normal model and computed probabilities from it.

The critical question we didn't ask: what if the data doesn't actually follow a Normal distribution?

Week 6 introduces two diagnostic tools:

- **Empirical CDF (ECDF):** sort the data and plot actual percentiles — no model required. Lay it over today's fitted curve to see where the model holds and where it drifts.
- **Q-Q plot:** plot observed quantiles against theoretical Normal quantiles. If the data is Normal, the points fall on a diagonal line. Curves, S-shapes, or outlier bends reveal departures from Normality.

Both of today's directions come back next week without the model — answered by sorting real records instead of fitting a curve.

Question for Week 6: The Normal model predicts the 99th percentile outdoor temperature in Tucson as 108°F. Counting directly from the ranked data gives 102°F. Why does the Normal model overpredict? What does the gap reveal about the distribution's actual shape?

Key Results — Week 5 Dataset Summary

Concrete f'c (psi):

Group	μ	σ	P(fail)	ACI status
All	4,496	442	13.1%	—
Alpha	4,632	344	3.3%	✓
Beta	4,367	484	22.4%	✗

10th percentile: Alpha = 4,191 psi ✓ · Beta = 3,747 psi ✗

Traffic speed (mph):

Group	μ	σ	85th percentile
All	54.4	7.8	62.5 mph
AM Peak	52.2	8.2	60.7 mph
PM Peak	57.2	6.2	63.6 mph

P(Speed > 45 mph): AM ≈ 81% · PM ≈ 99%

CostPerCY ($/CY): median=$185, mean=$196, skewness ≈ +1.5 → lognormal (σ_{ln}=0.35)

Week 5 In-Class Exercise — Normal Distribution in CE

`Week5_CE_Measurements.csv` · `Wk05.04_In_Class_Answer_Sheet.xlsx`

Dataset: 500 records · Supplier (B) · TimeOfDay (C) · fc_psi (D) · Speed_mph (E) · CostPerCY (F)

In-class target: Setup + Section A (concrete) + Section B1–B6 (traffic)

Section A — Concrete f'c (Normal distribution):

- A1–A4: AVERAGE, STDEV, MEDIAN, SKEW for all specimens and by Supplier
- A5–A8: NORM.DIST for P(fc < 4,000), P(4,200 < fc < 5,000)
- A9–A10: NORM.INV for ACI 318 compliance check (Alpha vs Beta)

Week 5 In-Class Exercise — Section B: Traffic + Cost

Section B — Traffic speeds (Normal probabilities):

- B1–B6: AVERAGE, STDEV, NORM.DIST for all observations and by TimeOfDay
- B7–B9: NORM.INV for 85th percentile speed — AM vs PM comparison
- B10–B12: NORM.DIST for P(Speed > 45 mph) posted speed compliance
- B13–B14: Lognormal preview — CostPerCY LN helper column

Charts and Memo (complete outside class if needed):

- Chart 1: fc_psi histogram · Chart 2: Speed_mph histogram · Chart 3: fc_psi vs Speed_mph scatter
- Memo: 3 paragraphs — concrete reliability · traffic compliance · interpretation of σ